

Ahmad Ibrahim

- Open to new roles

Cape Town, South Africa · Canadian citizen · remote

I design
data-heavy
product
interfaces,
and build
them.

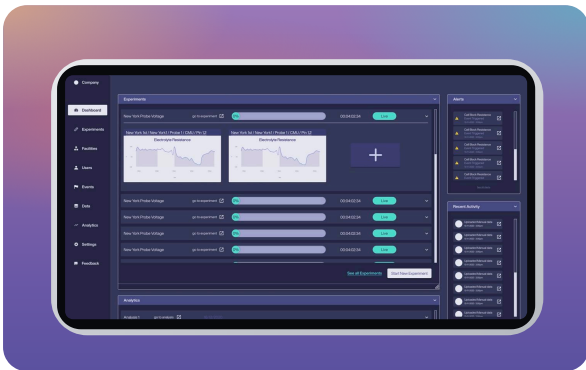
I design dashboards and internal tools for technical teams, then build them as working front ends.



- In use

Featured: Pulsenics System Software

Selected work



● In use

Pulsenics System Software

Client Pulsenics

Product Web app for EIS hardware

Role Product design and UI system, with Marc Varone

The app Pulsenics' clients use to manage their probes, run experiments and analyse the data. It's part of the company's core offering.

Product · Data viz · Design system



● Live

Gabion Developments

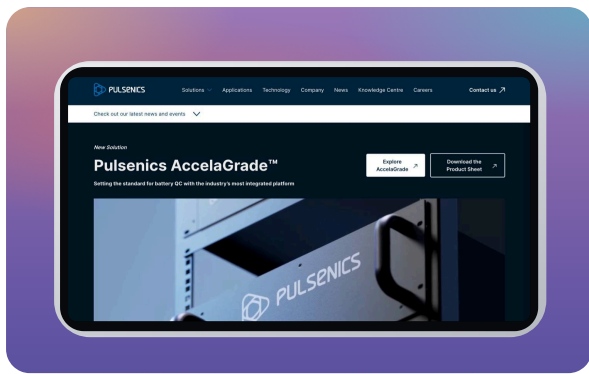
Client Toronto design-build firm

Product Marketing site

Role Sole designer and builder

A site that lets the firm's homes speak first. Designed from scratch and built in Webflow in two months.

Web · IA · Webflow



● Live

Pulsenics Website

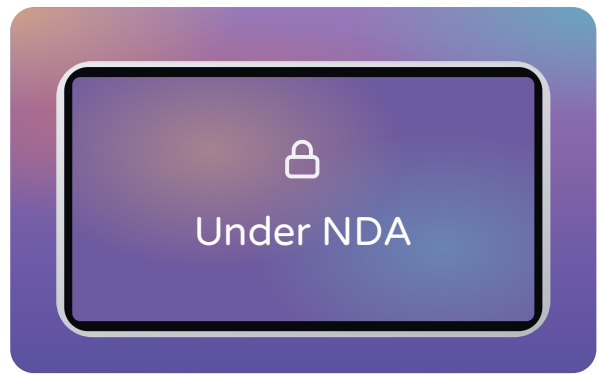
Client Pulsenics

Product Company site and Knowledge Centre

Role Design, build and upkeep

Ran a workshop with the marketing team that led to an Applications page, where visitors work out which product they need. Also built a Knowledge Centre for publishing what Pulsenics knows about EIS.

Web · IA · Webflow CMS



● In use

Under NDA

Profectus Group

Client Profectus Group

Product Internal data and reporting tools

Role Designed and built the front ends

Tools that replaced manual reporting. The team no longer logs in to every source and copies the data into spreadsheets, which saves each person an estimated 10–20 hours a week. Happy to walk you through it in an interview.

Product · Internal tools · SQL

About me

I started at Profectus Group in a data role. I learned everything I could about data and database management, ended up owning part of the data funnel, and then used my UX skills to make that work easier. My background is industrial design (OCAD University).

The thing I care about most in an interface: when you land on it, it should be clear what it's for and how to use it.

Focus

Data-heavy B2B products and internal tools

Design

Figma, Figma Make, design systems

Build

React, HTML/CSS, Webflow, Claude Code, Codex

Data

SQL and reporting

How I use AI. The same way I'd work without it. I write a brief, build out a design system, then work through the IA and the rest of the design. AI just makes that process faster.

Earlier work

Knowledge Leap

My OCAD University thesis: an open source AR app for Magic Leap that turns your surroundings into lessons.

[View →](#)

DuoVillage

A group research project: a Duolingo concept for practising real conversations without the social anxiety.

[View →](#)

Contact

Hiring for a product or UX role?

I'm open to roles on data-heavy B2B products and internal tools.

• In use

Product design

Web app

Design system

Pulsenics System Software

The web app Pulsenics and its clients use to manage their testing hardware, run experiments, control probes and analyse electrochemical data.

Overview

Before

Code sent straight to the probes

The team ran experiments from testbed software, sending commands to the hardware in code.

What we did

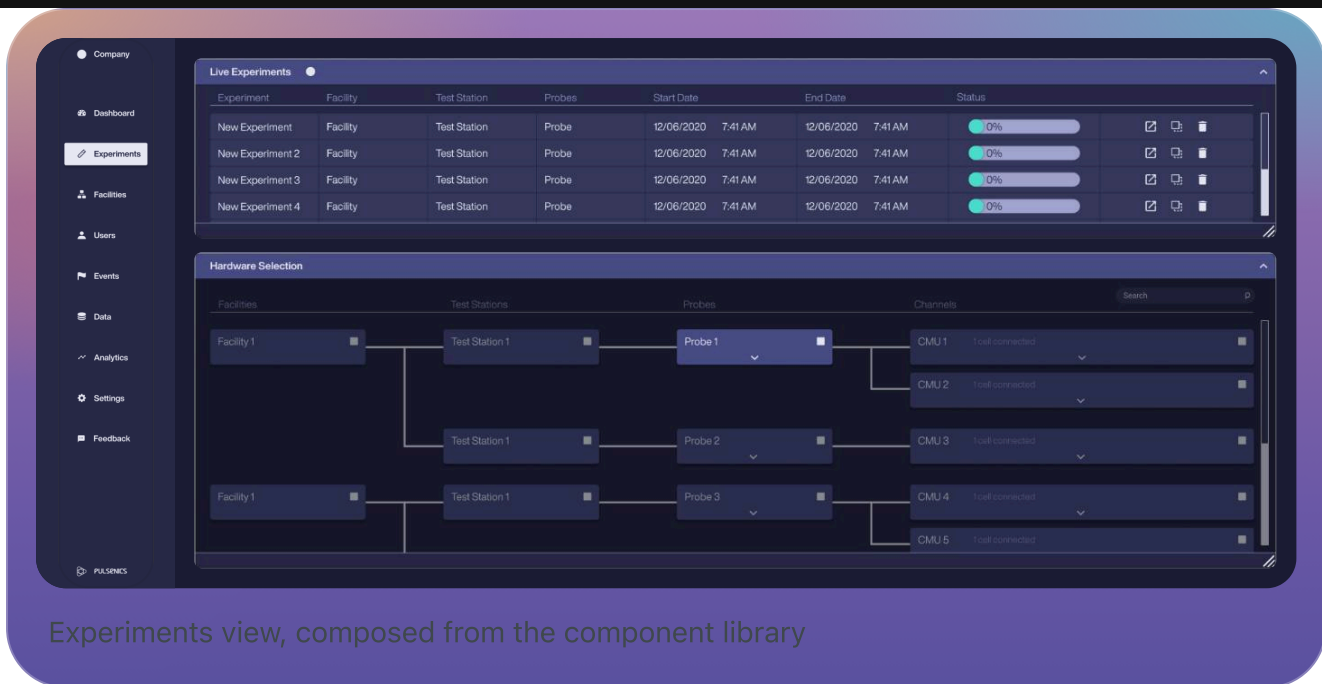
Made those steps visual

We met the engineers, mapped how their experiments need to run, and designed each part of the app and its UI system.

Result

Shipped with the product launch

Pulsenics' clients use it today as part of the company's core offering.



Client	Pulsenics, Toronto
Product	Web app for running experiments and managing probes
Team	Me and Marc Varone (50/50), with Pulsenics' backend team
My role	Product design and the UI system, from the ground up
Timeline	Designed from early 2021, shipped a year later
Status	In use

The brief

Design a tool that lets Pulsenics and its clients run experiments and manage every probe they have connected.

Pulsenics makes electrochemical impedance spectroscopy (EIS) hardware: probes that test the health and performance of batteries, fuel cells and electrolyzers without damaging them. The software sits between that hardware and the people using it. It

has to know what is plugged in where, start and stop measurements, and turn a stream of readings into something an engineer can act on.

Users

Technical teams

Engineers and researchers who know the chemistry and want precise control.

Data

Dense and continuous

Measurements stream in from many cells at once, across frequency and time, for hours or days.

Hardware

Physically nested

Facilities hold test stations, test stations hold probes, and probes connect to cells through channels.

How we got there

From commands in code to a visual UI

The app launched with the product. Until then, running an experiment meant using testbed software and sending commands to the probes in code. Our goal was to take that series of steps and make it visual.

1 Meet the engineers

We sat down with them and saw how they need an experiment to run.

2 Map the flow

We mapped it from plugging in hardware to analysing the results. That map is below.

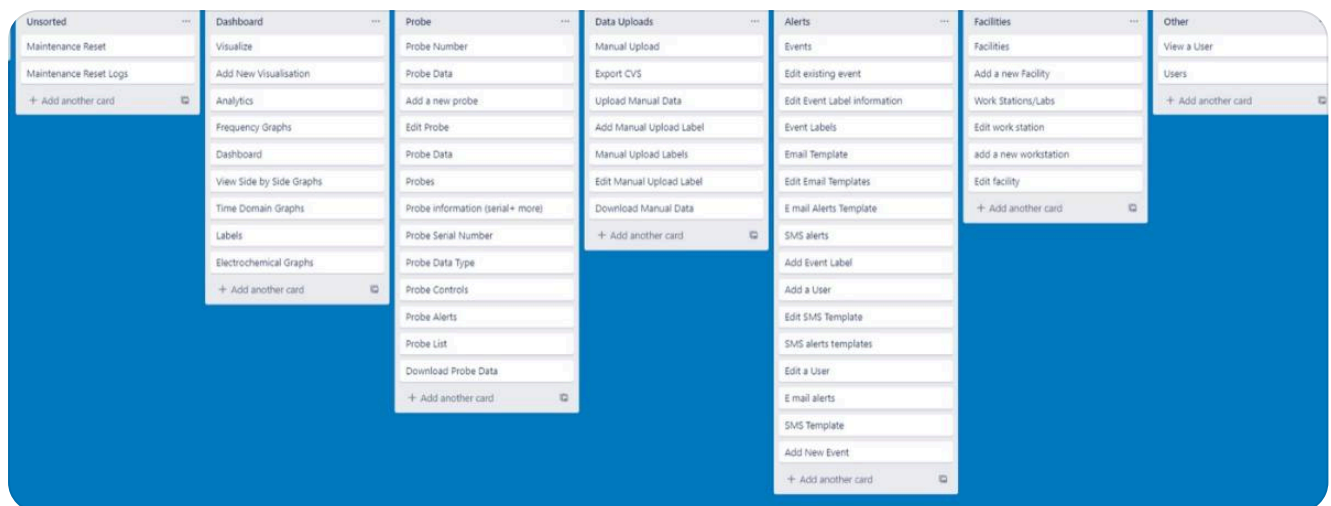
3 Build it section by section

Then we designed each part in turn, from wireframes to final UI.

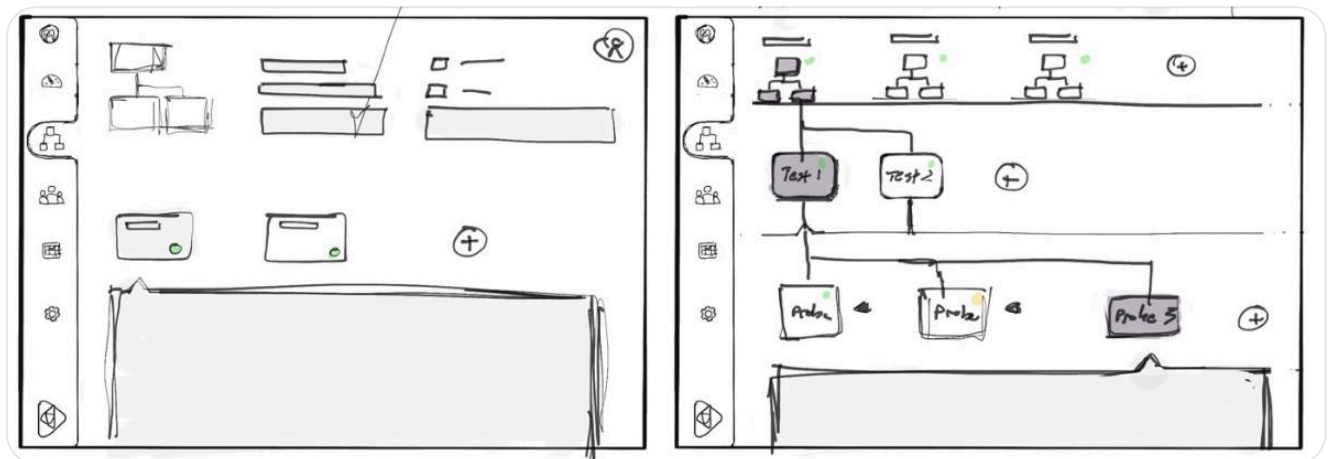
We worked closely with the backend team the whole way, so we understood quickly what was and wasn't possible. There was pushback on both sides at times.

The Figma file still shows the path: an open card sort with Pulsenics staff and user flows, sketches, eight rounds of wireframes, a foundation for the UI system, then the

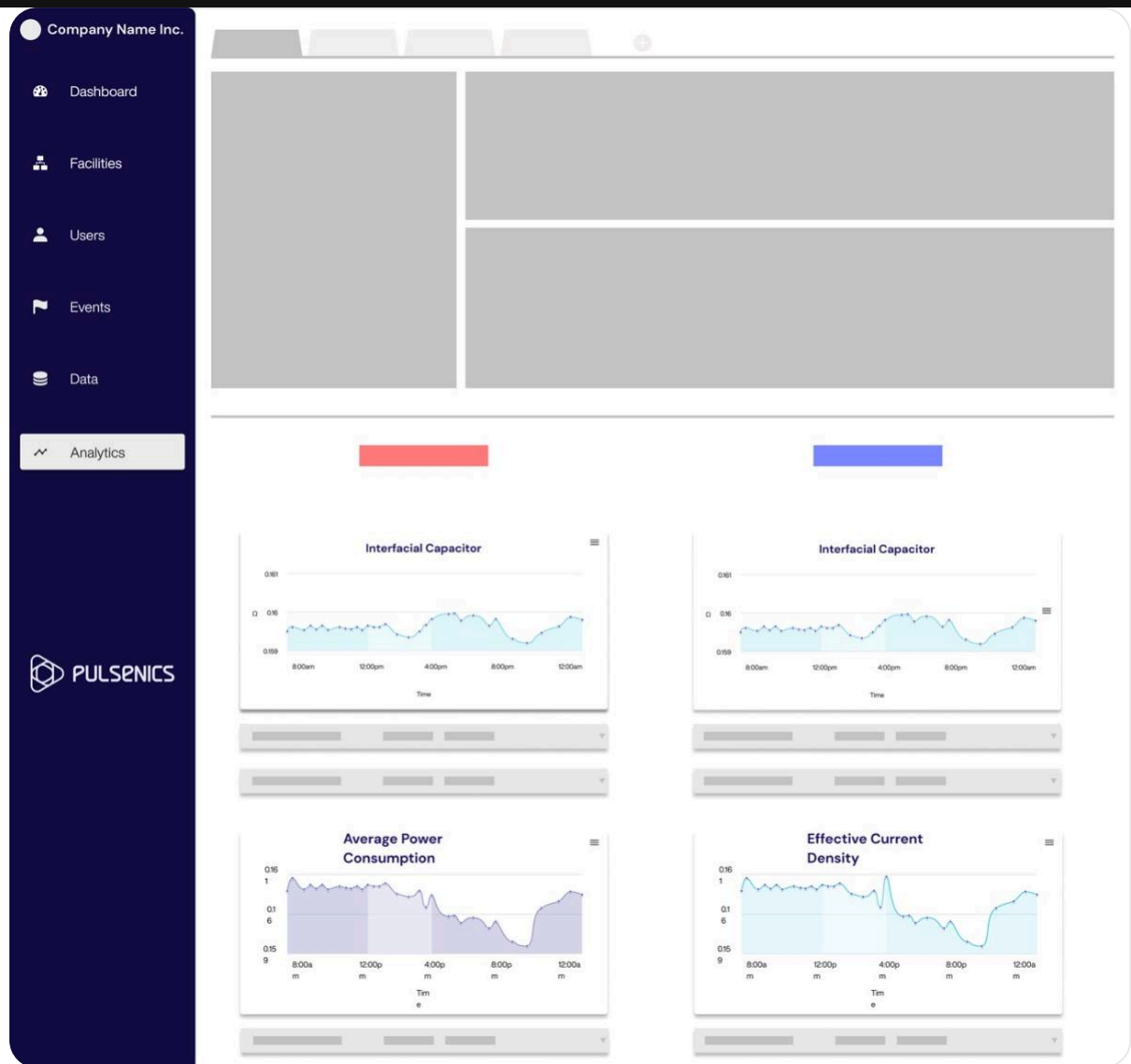
final screens and handoff to developers.



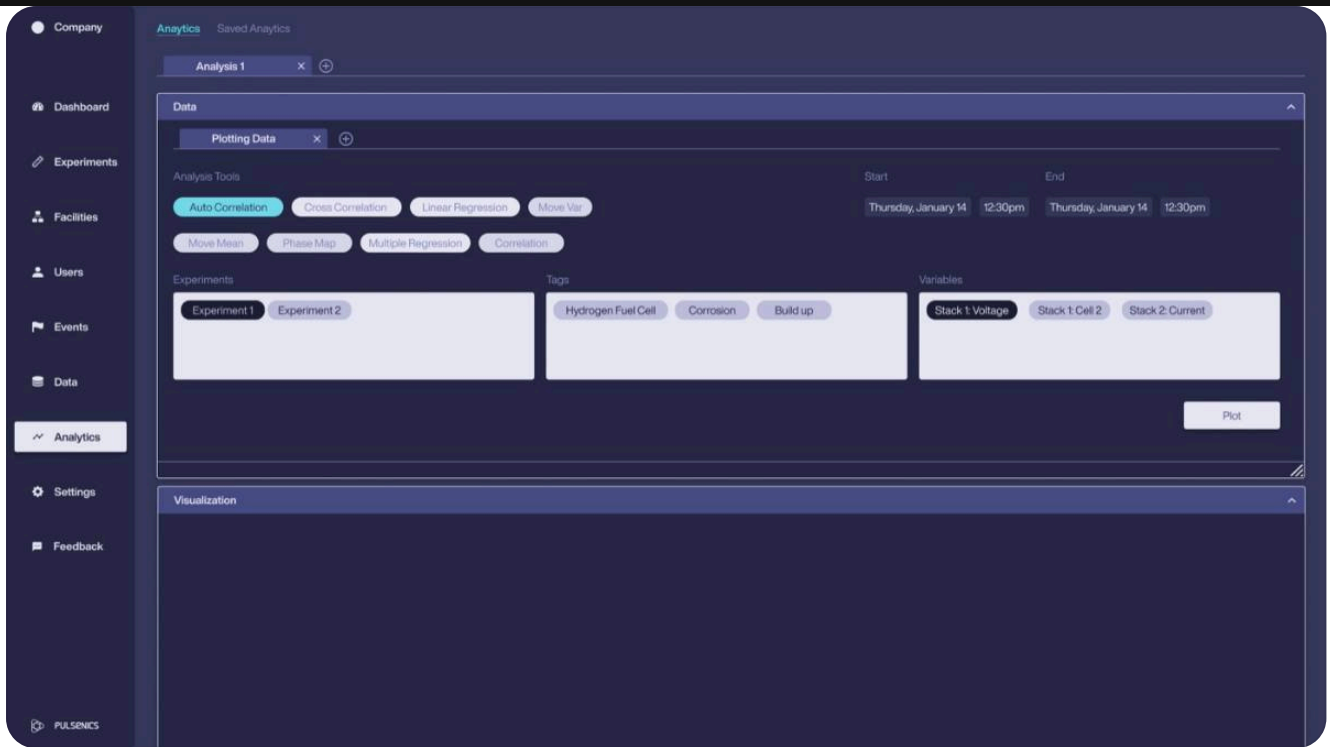
An open card sort with Pulsenics staff, from the CEO and engineers to the marketing team



Sketches by [Marc Varone](#). The hardware tree on the right became decision 01.



Wireframe V1: analytics as a page of fixed charts



Wireframe V8: analysis built from experiments, tags and variables. The final version is decision 05.

One flow, end to end

This is the flow we mapped with the engineers. Everything in the app supports it, from plugging in hardware to explaining what the data shows.

01

Pick the hardware

02

Start the experiment

03

Control the probe

04

Watch it live

05

Mark what happened

06

Analyse the results

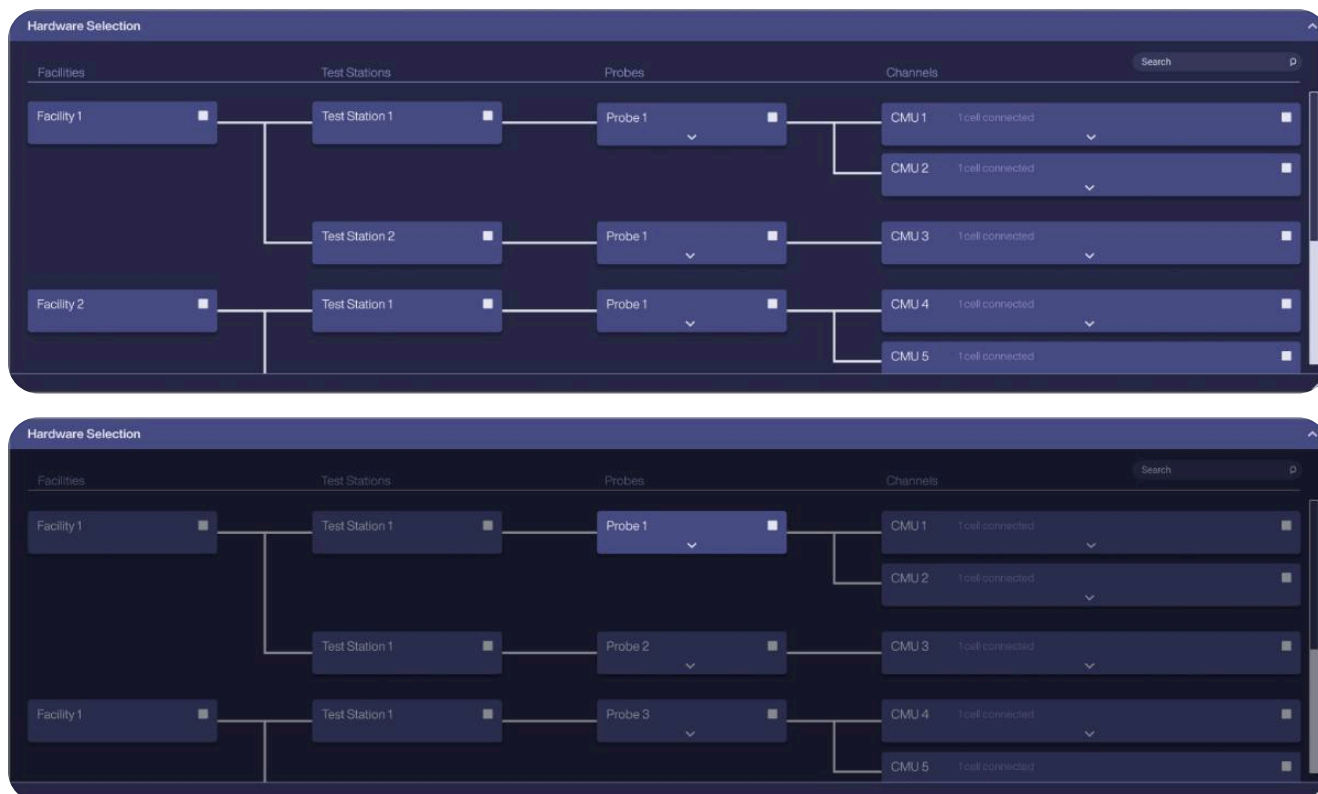
Key design decisions

01

Mirror the physical setup

Hardware is chosen from a connected tree that matches how the equipment is actually wired: Facilities, then Test Stations, then Probes, then Channels, with each channel showing how many cells are connected.

Selecting any node dims every branch it doesn't belong to, so it's always clear which cells an experiment will touch. Search sits on top for setups too large to scan.



The full tree, and the focus state when a probe is selected

02

Make starting an experiment one decision

Once hardware is picked, starting is a small pop-up: a start and end time, with two shortcuts for the most common cases, *Start Now* and *No specified end time*.

Start Experiment

⌚

Start

Thursday, January 14

12:30pm

☐

Start Now

End

Friday, January 14

12:30pm

☐

No specified end time

Start Experiment

Start Experiment

03

Give direct control of the instrument

Probe controls expose what the hardware actually does: choose a probe, set the time between samples, switch between galvanostatic and potentiostatic mode, and enter up to ten frequency and amplitude pairs.

Send Controls and a red *Stop Sampling* sit together at the end, so the safe way out is always visible. Tabs keep the controls history, alerts and maintenance logs one click away.

Current ControlsControls HistoryAlertsMaintenance Reset Logs

Time Between Samples (s): 0

Mode: ☒ Galvanostatic ☐ Potentiostatic

Number	Frequency (Hz)	Amplitude (A)	Reset
1	800	0.1	Reset
2			Reset
3			Reset
4			Reset
5			Reset
6			Reset
7			Reset
8			Reset
9			Reset
10			Reset

Send ControlsStop Sampling

Probe controls

04

Keep running work in view

Live experiments get their own table with facility, test station, probe, timing and a progress bar, plus quick actions to open, duplicate or delete. Once a run finishes, it moves to Saved Experiments, so the live table only shows what's actually running.

Live Experiments

Experiment	Facility	Test Station	Probes	Start Date		End Date		Status			
New Experiment	Facility	Test Station	Probe	12/06/2020	7:41 AM	12/06/2020	7:41 AM	0%			
New Experiment 2	Facility	Test Station	Probe	12/06/2020	7:41 AM	12/06/2020	7:41 AM	0%			
New Experiment 3	Facility	Test Station	Probe	12/06/2020	7:41 AM	12/06/2020	7:41 AM	0%			
New Experiment 4	Facility	Test Station	Probe	12/06/2020	7:41 AM	12/06/2020	7:41 AM	0%			

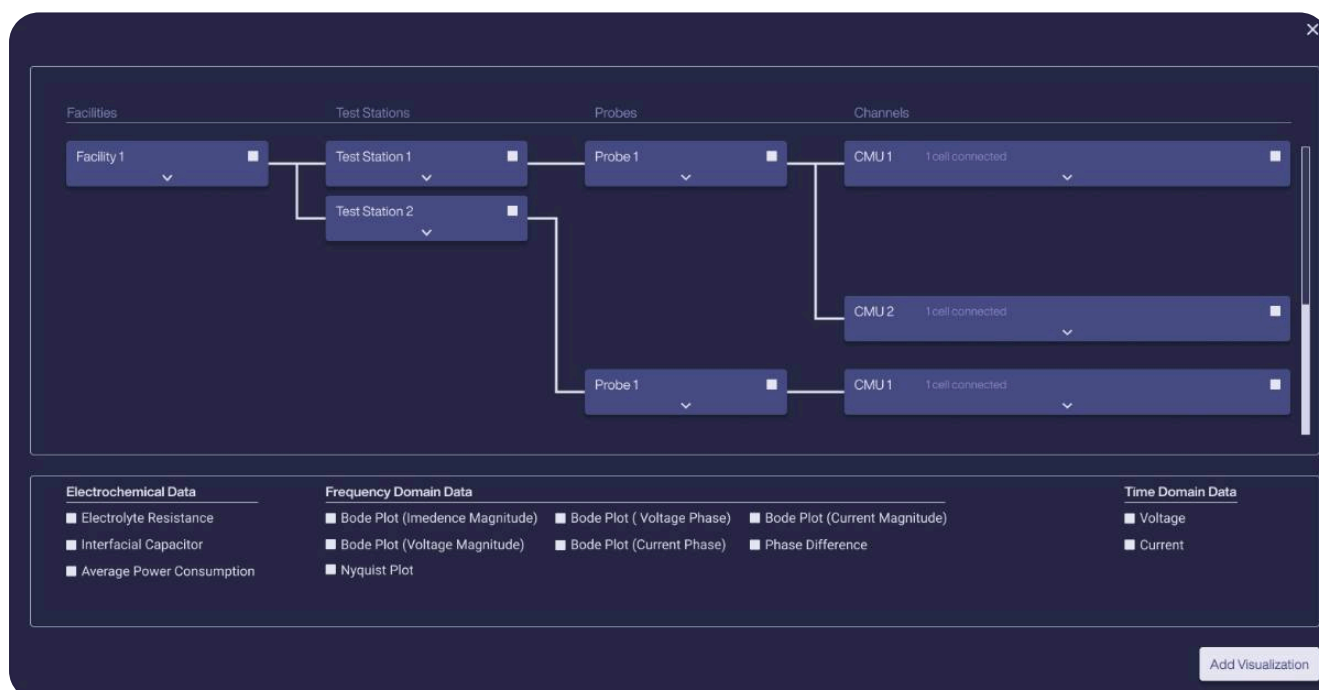
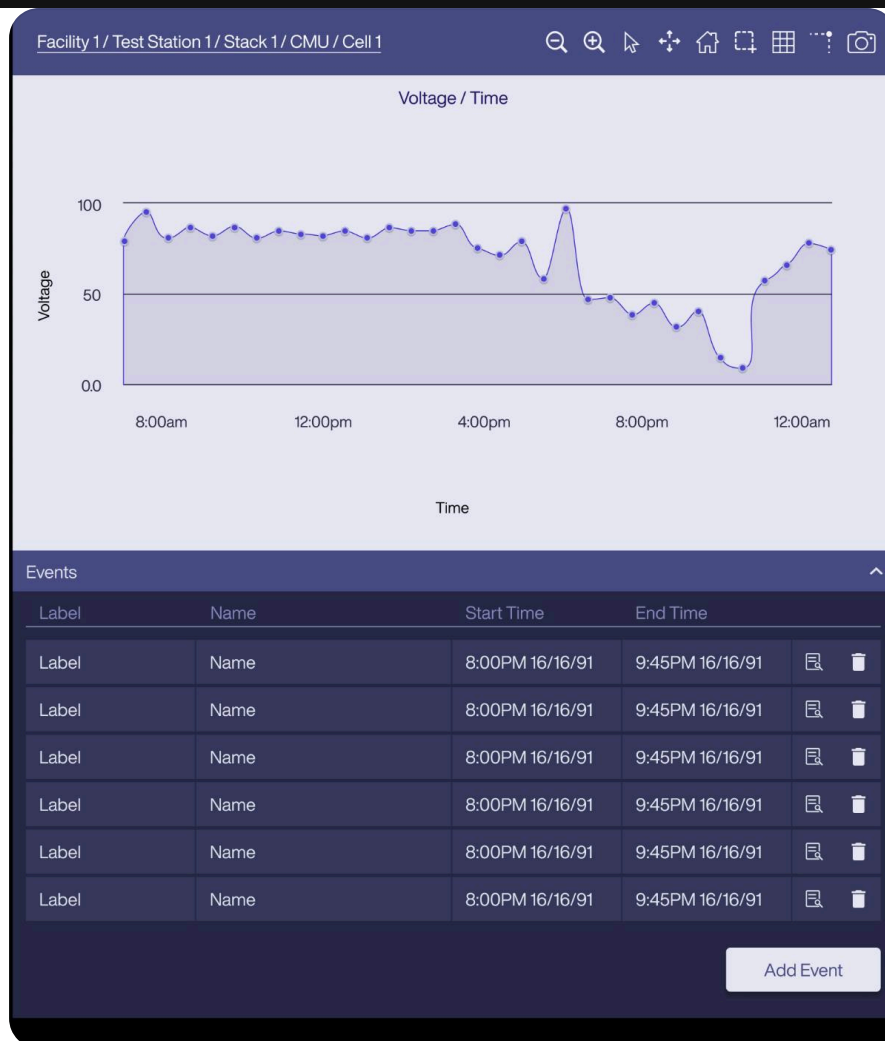
Live experiments

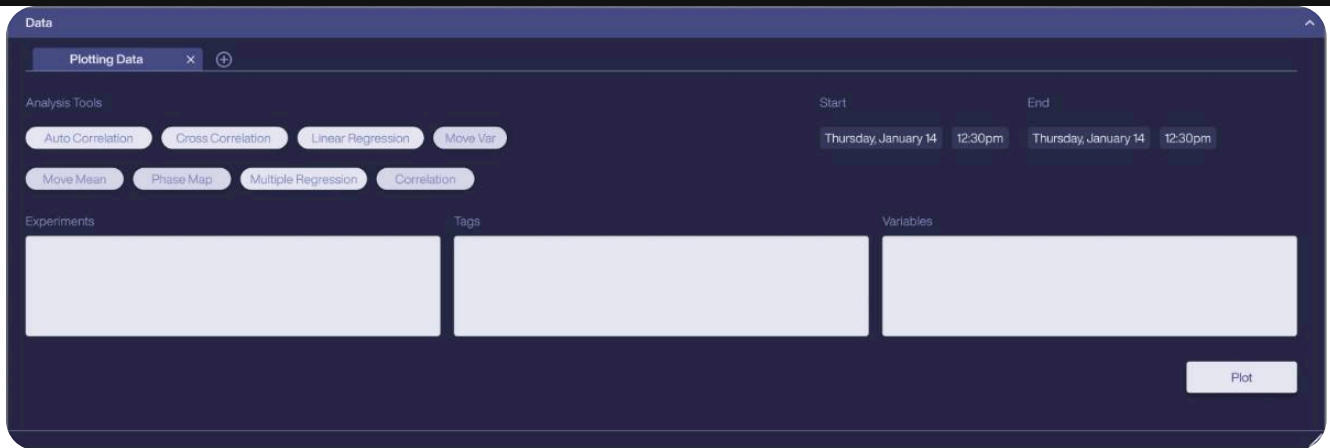
05

Work with the data in one place

Every graph shows its full hardware path at the top. You always know which cell you're looking at. If something odd shows up, you can select that range and save it as a tagged *Event*. The whole team can find it later.

Adding a visualization reuses the hardware tree, then groups the data the way electrochemists think about it: electrochemical, frequency domain and time domain. Analysis tools such as correlation, regression and moving averages sit in the same app, each opening in its own tab.





Events on a graph, adding a visualization, and the analysis panel

Built as a system

Marc and I built the design system alongside the screens. It gave Pulsenics a set of reusable assets and made prototyping much easier.

Sections

Collapsible, resizable panels

Every area of the app (hardware, tags, live and saved experiments, data, visualizations) is a panel that can be opened, closed or resized. Users set the workspace up around what they're working on.

Navigation

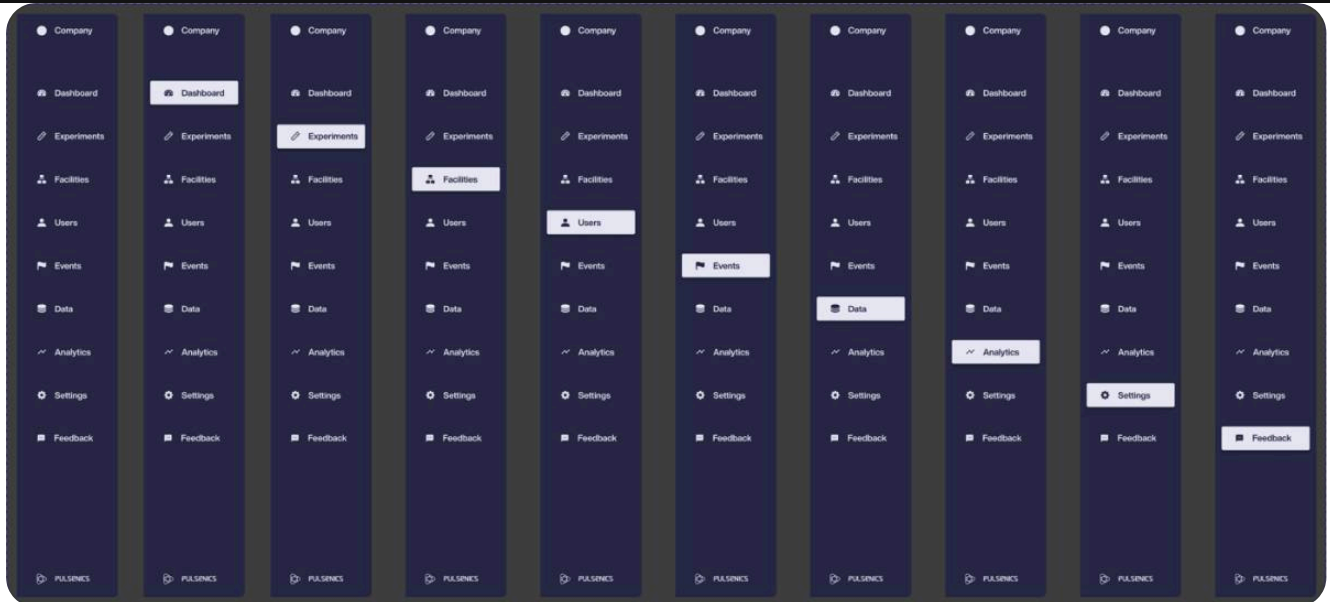
Nine sections, two widths

Dashboard, Experiments, Facilities, Users, Events, Data, Analytics, Settings and Feedback, each with a selected state, plus a slimmer rail with just the icons for more room.

Charts

One graph component

The same graph, toolbar and events tab are used everywhere data is plotted, in four states from no events to many.

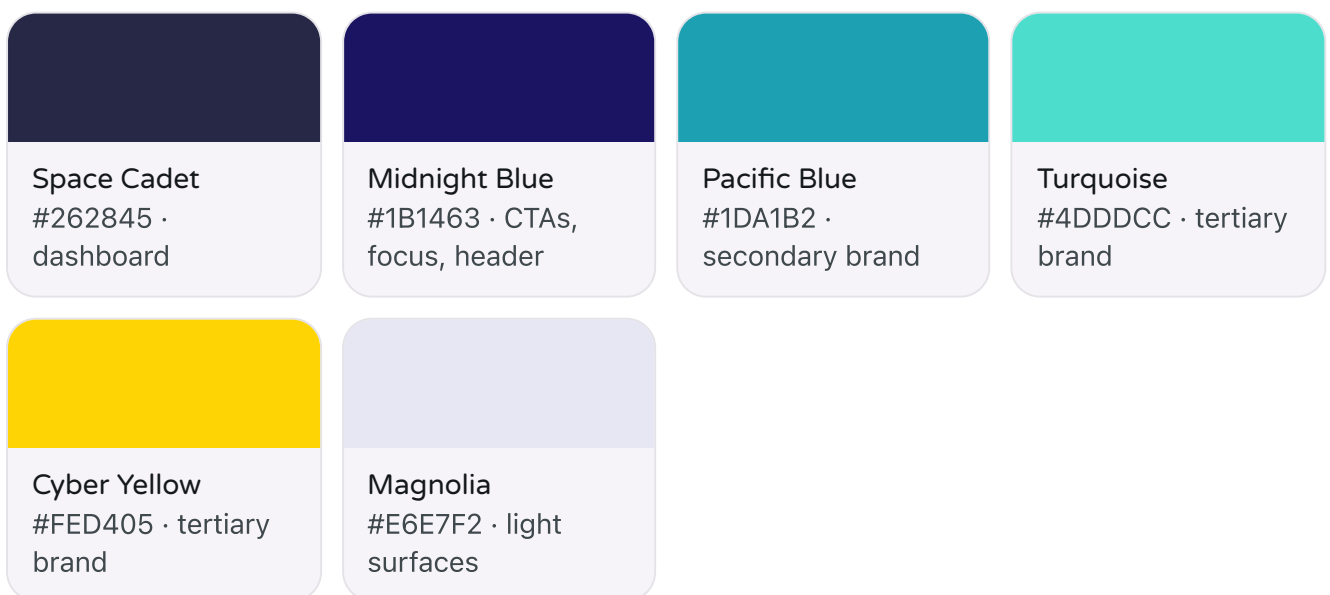


Navigation, one state per section



Part of the component library: buttons, tabs and analysis chips

Colour



What I'd refine next

Contrast

Secondary text and column headers sit close to the panel colour. I'd raise them to meet WCAG AA, which matters for people staring at data all day.

Inputs on dark

The light input fields are the brightest thing on screen. A darker field with a clear focus ring would keep attention on the data.

Hierarchy

Most panels carry the same visual weight. Giving live data more presence than setup controls would help scanning.

Credits

Designed with Marc Varone

Pulsenics System Software was designed by two of us. See more of Marc's work at marcvarone.com.

Gabion Developments

A marketing site for a Toronto design-build firm. It puts their finished homes up front and makes booking a call the obvious next step.

Overview

Brief

Their work, front and centre

A site in the firm's own style that markets its homes and gives people a professional way to reach them.

What I did

Designed and built it alone

Structure, visual design and a responsive Webflow build, in two months, with the owners and their marketing lead.

Result

Launched in 2022

The client loved it. It looked like their work, and the homes do the selling.



Homepage, desktop and mobile

Client	Gabion Developments, Toronto
My role	Sole designer and site builder
Platform	Webflow
Timeline	2 months, concept to launch
Launched	2022
Scope	IA, UX/UI, visual design, responsive build

The brief

Gabion needed a site to market its home developments, and to give prospective clients a professional way to find them.

They wanted a site that spoke to their own design style, one that let their work speak first. That's what I led with: a video of a finished home on the homepage, and minimal styling so the photos and video they provided carry the page.

What the site had to do

Sell with finished work

People judge a builder by what it has already built, so the site had to put real homes in front of visitors straight away.

Make the process feel manageable

Building or renovating a home is a big, unfamiliar decision. The site needed to show what happens, and in what order, before anyone picks up the phone.

Turn interest into a conversation

One clear next step, repeated across the site: book a call.

How we worked

Two months, with feedback the whole way through

We started with a kickoff with the owners. After that I worked with their marketing lead on the rest of the site, with the owners checking in along the way. They gave feedback continuously and supplied the content: the photos and videos of their homes.

1 Structure

Mapped the seven pages and the path from a first visit to Book a Call.

2 Design

Set the typography, grid and image treatments so the photography leads.

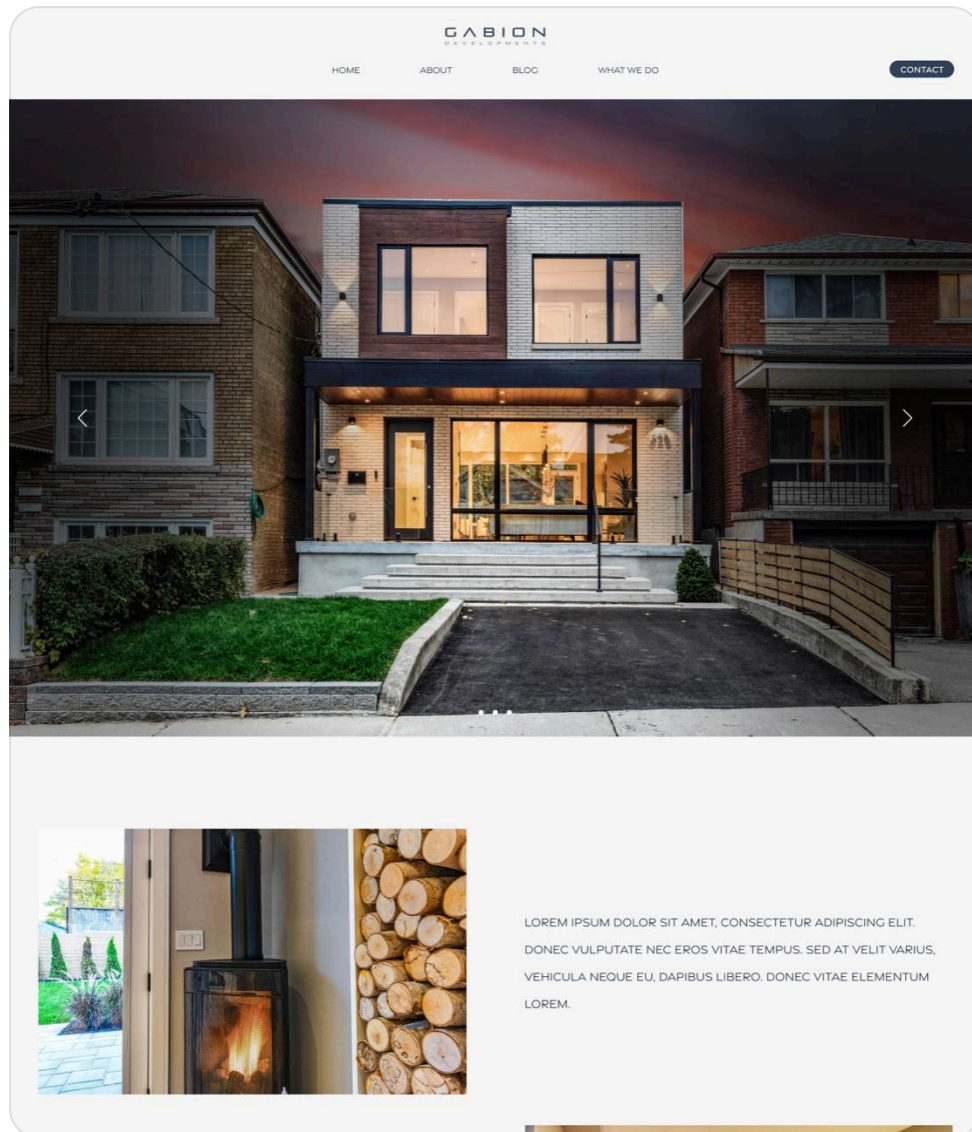
3 Build

Built responsive pages in Webflow from reusable sections, so the firm's team can edit copy and add projects without a developer.

4 Launch

Checked every page across screen sizes, then went live.

Early concepts, before the final direction:



Home v1 opened on a photo carousel. The live site moved to a video of a finished home.

WHAT SHOULD WE DEVELOP TOGETHER?

LOREM IPSUM DOLOR SIT AMET, CONSECTETUR
ADIPISCING ELIT. DONEC VULPUTATE NEC EROS VITAE
TEMPUS. SED AT VELIT VARIUS, VEHICULA NEQUE EU,
DAPIBUS LIBERO. DONEC VITAE ELEMENTUM LOREM.

HOW WOULD YOU LIKE US TO CONTACT YOU?

☐ CALL

☐ EMAIL

WHAT SHOULD WE DEVELOP TOGETHER?

LOREM IPSUM DOLOR SIT AMET, CONSECTETUR
ADIPISCING ELIT. DONEC VULPUTATE NEC EROS VITAE
TEMPUS. SED AT VELIT VARIUS, VEHICULA NEQUE EU,
DAPIBUS LIBERO. DONEC VITAE ELEMENTUM LOREM.

CONTACT US

HOW WOULD YOU LIKE US TO CONTACT YOU?

☐ CALL

☐ EMAIL

Information architecture

A flat structure with seven pages

The only nested level is the individual project pages, so nothing sits more than two clicks from the homepage.

Home	Video hero, welcome, philosophy, Book a Call banner
Work With Us	The journey, in four stages
Projects	Selected work and upcoming projects (below)
Testimonials	Homeowner quotes; pick a name to read each one
Blog	Image gallery linking to Instagram
Careers	Join Our Team, with a link to current openings
Contact	Enquiry form with a "How did you hear about us?" field

Projects: Selected work

Finished projects, each with its own page.

Cedarvale Upper Beaches
Bayview

Projects: Upcoming

Projects in the pipeline, shown as architectural drawings.

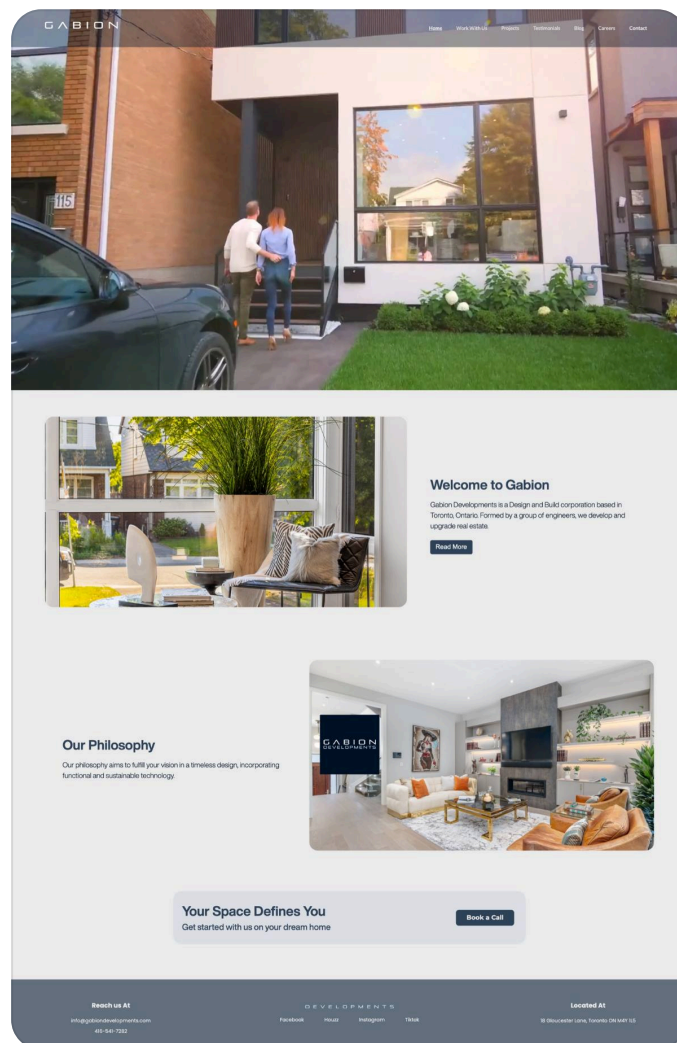
Laneway Trinity Bellwoods
Beaches Oakwood
East York Port Credit
Port Union

Six design decisions

01

Open with a finished home

The homepage opens on a video of a completed project, 111 Malvern Ave, that fills the whole screen. There's very little copy. Before any copy about the company, you see one of their homes. The welcome and philosophy sections come after that.

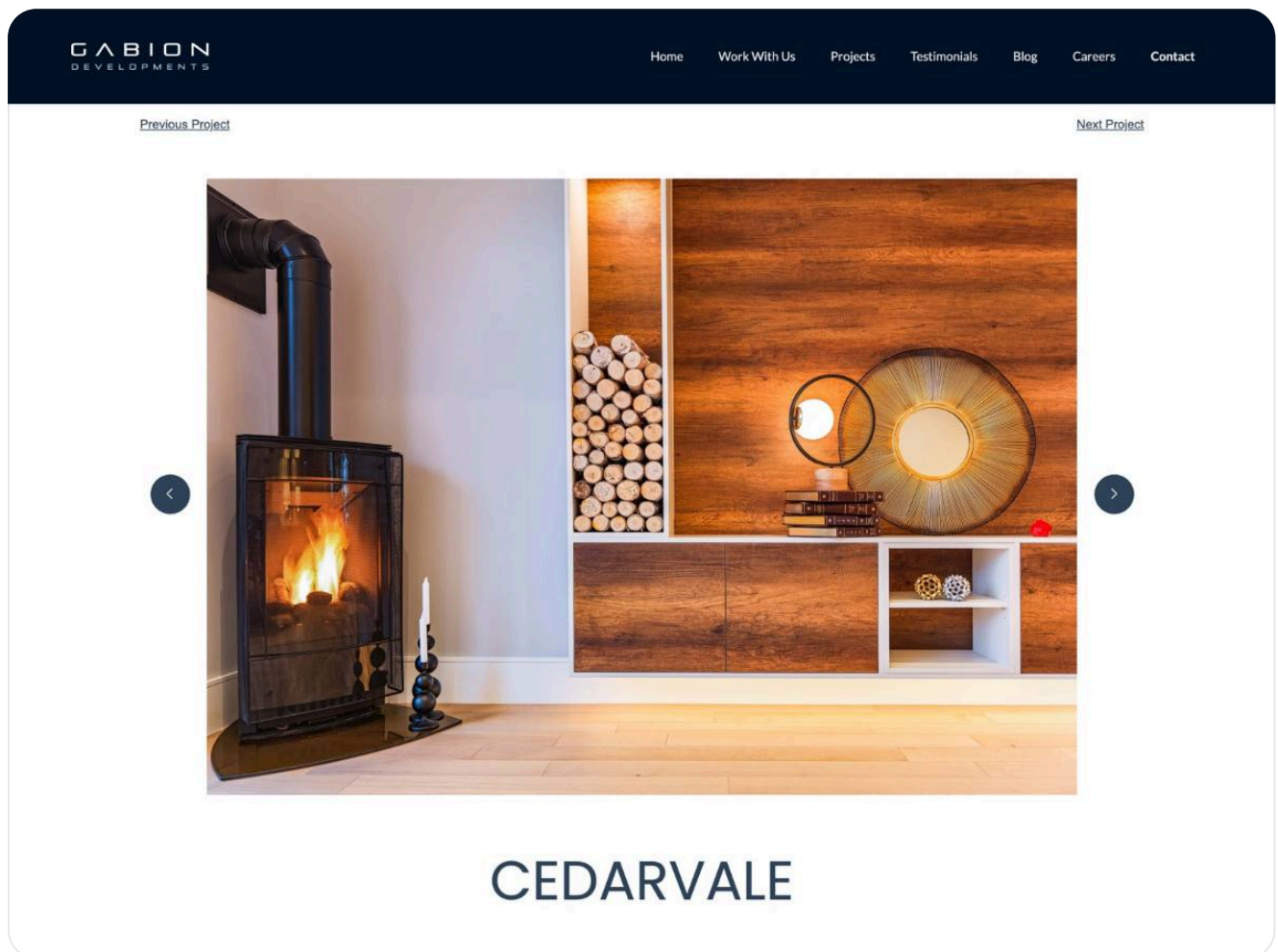


Homepage, full length

02

Let the photography carry the design

I kept the interface simple so the photos could do the work. Every page is built around large images, with plenty of white space and very little else competing for attention. Each project page leads with a photo carousel, keeps the description short, and links to the previous and next project so visitors can keep browsing.



Project page: Cedarvale

03

Separate what is built from what is coming

The Projects page splits Selected Work from Upcoming Projects. Finished homes are shown in photographs and each gets its own page. Upcoming ones are shown as architectural drawings, so the difference is clear at a glance. Visitors can see what's finished, and that there's more on the way.

Selected Work

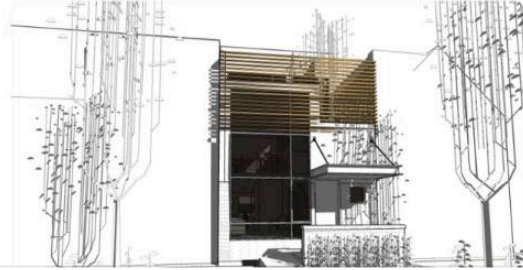


Cedarvale



Upper Beaches

Upcoming Projects



Projects: photos for finished work, drawings for what is coming

04

Show the process as four stages

Work With Us walks through Exploring, Imagining, Creating and Finalizing. Each stage explains what Gabion does at that point, such as visualizing design options and submitting the permit while Imagining, or monitoring construction while Creating. The images move from a pencil sketch to a rendering, a framing photo and the finished house, so the page shows the project taking shape.

It shows a new client what happens after they get in touch.



Exploring

During this phase, we dream up the vision for your space. We take the time to learn what is important to you and how you picture life in your new home.

Imagining

It's time to put pen to paper. Here is when we take your vision and visualize your space using architectural alternatives. This is also where we engineer conscious innovation into your home, by supplying you with options for cost and waste reduction technologies that will improve your quality of life. We continue to refine these designs until we have a formal plan with an estimated budget and timeline. At this stage, we also submit a permit to the municipality for approval.

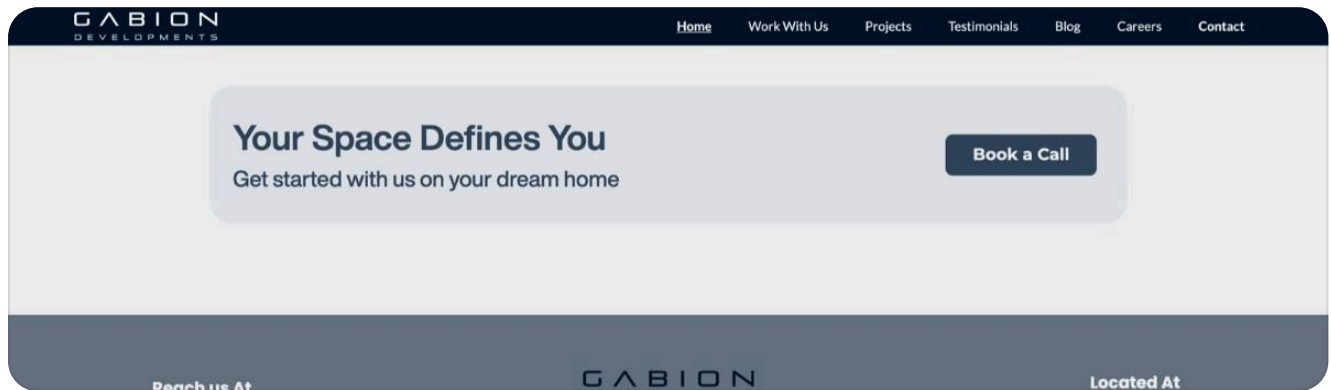


Work With Us: the journey, from sketch to site

05

One call to action, repeated

Book a Call is the main way to start. A banner reading “Your Space Defines You” closes almost every page, from the homepage to each project and the contact page, so there's always a way to get in touch close by. The footer below it repeats the email, phone number and address.



The Book a Call banner and footer, repeated across the site

06

Proof from clients, and a blog that runs on Instagram

Testimonials gives homeowners' own words their own page: one quote at a time on desktop, chosen from a row of names, and the full list of quotes stacked on mobile. The Blog is a gallery of image tiles that link out to the firm's Instagram posts, so the team only has to post once and the site stays up to date.

"I trust Gabion with the details, starting from the flow of the space all the way down to the placement of the kitchen sink and the lights. I came to them with a project to renovate my small duplex, and somehow, they expanded my space so much with the right design elements."

Maria M

Toronto home owner

Anna Y

Toronto home owner

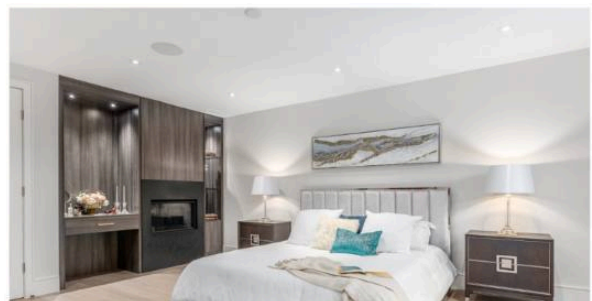
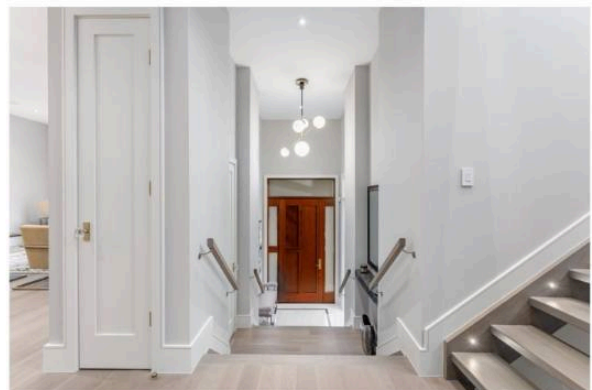
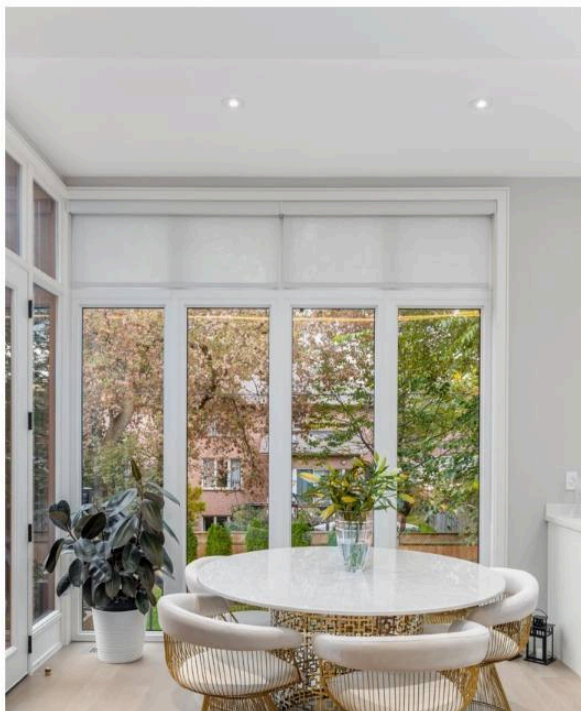
Melissa K

Toronto home owner

Karine G

Toronto home owner

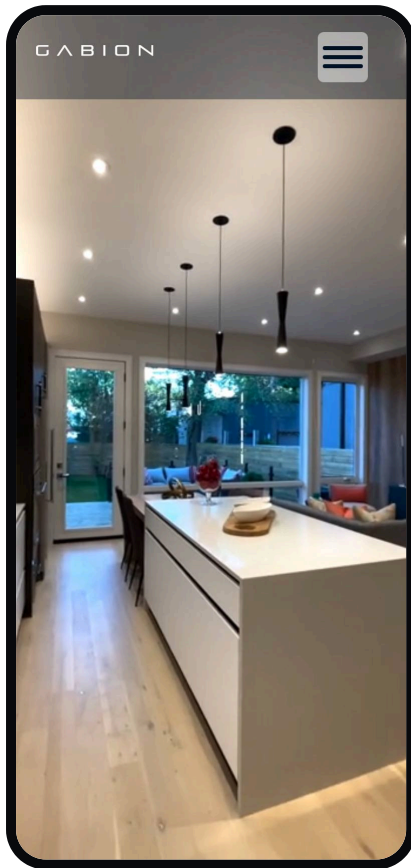
Blog



Testimonials and Blog

The same site on a phone

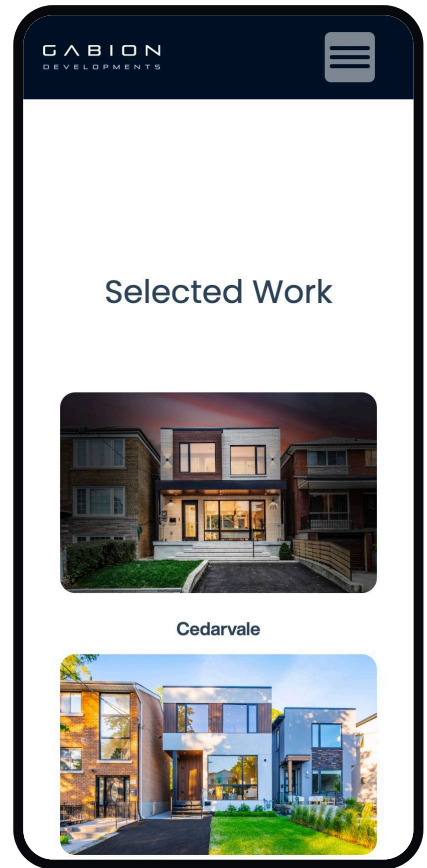
The navigation collapses behind a menu button, paired layouts stack image over text, and grids drop to one column. Components that rely on a wide screen change form: the testimonial selector becomes a simple list of every quote.



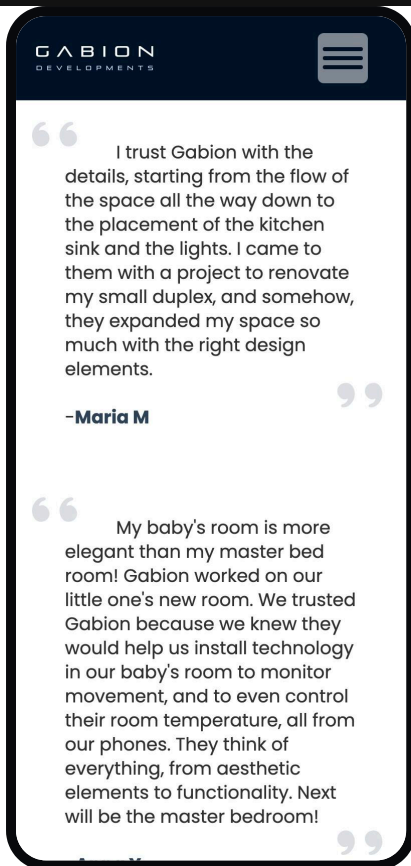
Home. The menu folds into one button.



Work With Us. Each stage stacks its image above the text.



Projects. Two columns drop to one.



Testimonials. The name selector becomes a list of quotes.

The result

Launched in 2022, and the client loved it

It looked like their work, which was the whole point. I handed it over after launch and don't maintain it now.

[Visit gabiondevelopments.com ↗](https://gabiondevelopments.com)

Pulsenics Website

I've worked on pulsenics.com since 2020. When Pulsenics wanted to add a lot of new pages, Marc Varone and I pushed for a workshop with their marketing team first. The Applications page came out of it, and it's now how visitors work out which product is right for them. I also built the Knowledge Centre, to improve the way Pulsenics publishes what it knows.

Overview

Problem

A lot of new pages to add

Before building them, we needed to know what a potential client should see when they land on the site.

What I did

A workshop, an Applications page and a Knowledge Centre

Marc and I ran a workshop with marketing that led to the Applications page. I also built the Knowledge Centre and set the site up on the CMS so staff without a technical background can publish.

Result

The main way clients find Pulsenics

The team updates it constantly with news and Knowledge Centre content, and it's the primary way potential clients find the company.

Check out our latest news and events

New Solution

Pulsenics AccelaGrade™

Setting the standard for battery QC with the industry's most integrated platform

Explore
AccelaGrade

Download the
Product Sheet



pulsenics.com homepage, desktop and mobile

Client	Pulsenics, Toronto
My role	Design, build and upkeep since 2020
Platform	Webflow CMS
Editing	Set up so people without a technical background can publish their own content
Scope	Information architecture, homepage and product pages (with Marc Varone), Knowledge Centre, News
Status	Live, and I still maintain it

The challenge

A technical product, for scientists and engineers

Pulsenics sells electrochemical impedance spectroscopy (EIS) hardware to battery, hydrogen and fuel cell teams. Its buyers are scientists and engineers. The site has to explain the products and keep up with a steady stream of news and studies.

I inherited the site from its original designer and kept its established look. Since then most of the pages have been redone or added as the company launched new products and needed new ways to explain its technology.

Information architecture

With Marc Varone

Help each visitor find the product they need

Pulsenics wanted to add a lot of new pages, and a new homepage hero. Rather than tack them on, Marc and I pushed for a workshop with their marketing team around one question: when a potential client lands on the site, what should they see, and how do we help them? The answer was that people didn't understand what each product was for. So we planned the site around two kinds of visitor.

Two ways in

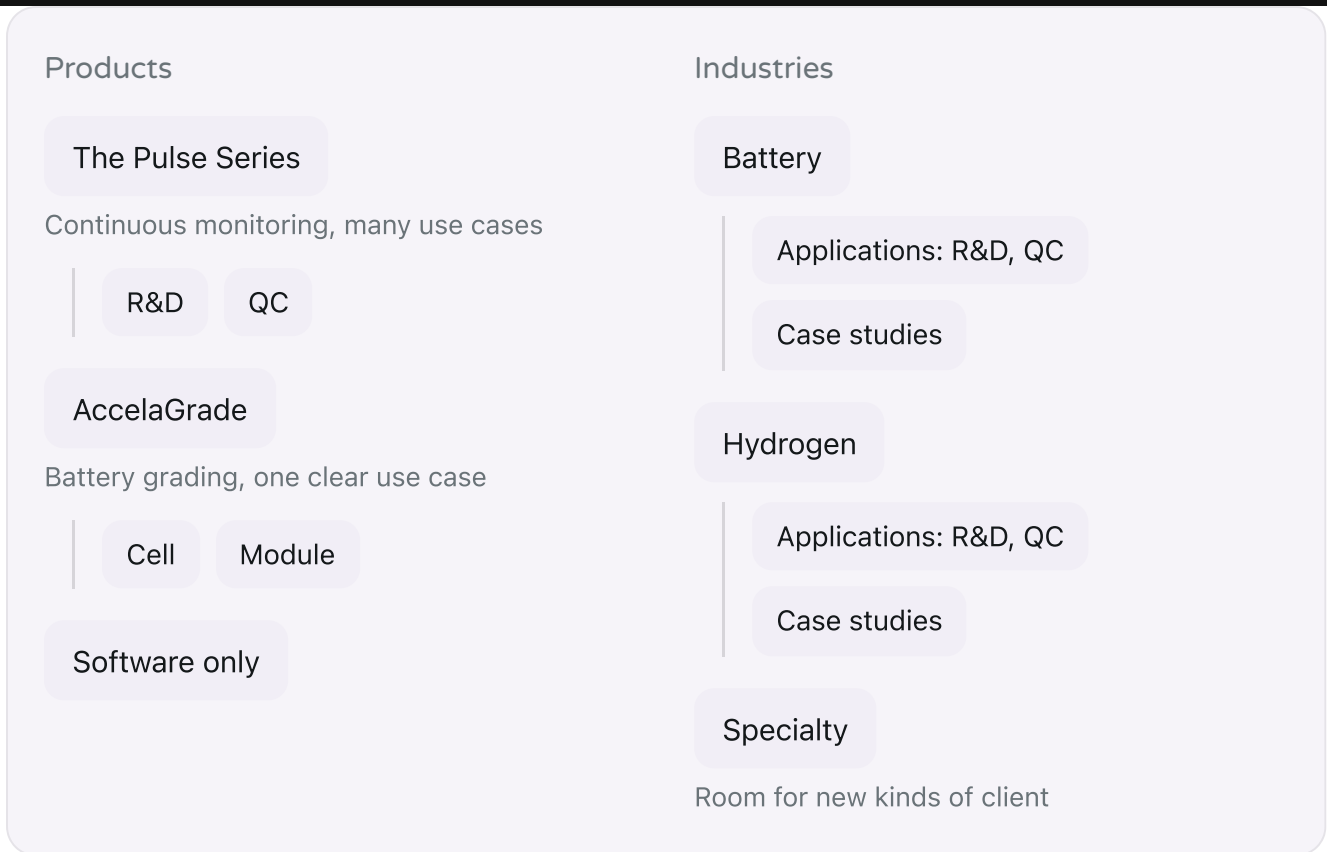
Knows Pulsenics → Solutions or Technology → Product

New to Pulsenics → *"I have a problem, I need to find the solution"* →

Applications → Product

From the workshop: the people who already know the products, and the people who only know their problem.

From there we built two trees: one for the products and one for the industries they serve. Every industry splits into R&D and QC work, with room for case studies as Pulsenics publishes them.



Then we mapped every application to the person who would use the product, the person who would buy it, and the product that fits.

Application	End user	Buyer	Product
Factory cell and module QC	Test engineers	Head of QC, NPI managers	AccelaGrade
Remanufacturing QC	Process engineers	Head of factory operations	AccelaGrade
Cell and stack R&D	Research scientists	Head of R&D, Head of Technology	The Pulse Series

A few rows from the map. The full version covers battery, hydrogen and specialty work.

That map became the Applications page, which was our idea. It's where visitors decide which product is right for them. Each application points to one product, so nobody has to work out the product range on their own.

Battery

Module qualification · Second-life
EV packs

AccelaGrade Module

Battery

Incoming cell QC · Remanufacturing
for motive and micromobility

AccelaGrade Cell

Battery

Degradation benchmarking and
diagnosis

The Pulse Series

Hydrogen

Stack failure analysis · Degradation
benchmarking · Electrolyser stack
defects

The Pulse Series

Homepage and product pages

With Marc Varone

01

Lead with the newest product

The homepage opens on AccelaGrade, the company's new battery grading system, with two clear actions: explore the product or download the spec sheet. It's their latest product, so it gets the top spot.

Check out our latest news and events ▾

New Solution

Pulsenics AccelaGrade™

Setting the standard for battery QC with the industry's most integrated platform

Explore
AccelaGrade ↗Download the
Product Sheet ↗

02

Two product lines, compared at a glance

AccelaGrade and the Pulse Series sit side by side in matching cards: a short line on what it's for, four capabilities each, and a single Learn More link. It's easy to see which one you need.

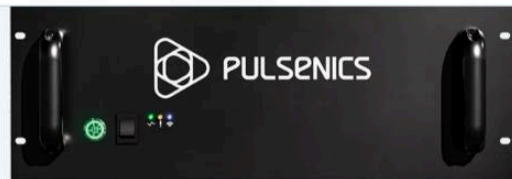


The Pulsenics AccelaGrade™

Grade battery health rapidly; built for cell and module QA/QC, remanufacturing, and repurposing.

- ✓ High Throughput Screening
- ✓ Ability to Test Every Cell
- ✓ Capacity Grading and Sorting
- ✓ Cell Matching for Second Life Use

Learn More ↗



The Pulse Series

Characterize the performance of your cells, stacks, modules and packs, in the lab or in your gigafactory.

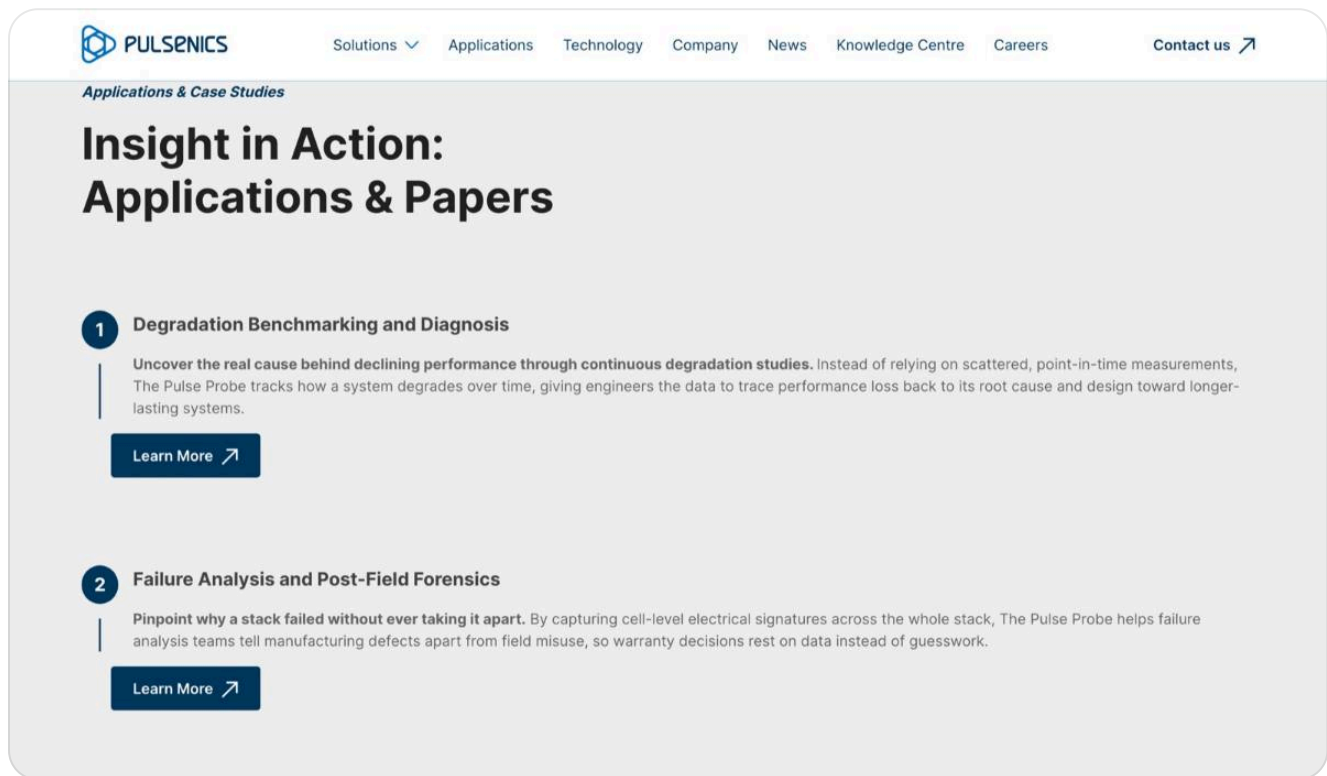
- ✓ Simultaneous Multi-Cell Measurement
- ✓ Rapid Multi-Sine Excitations
- ✓ Accuracy Across a Wide Frequency Bandwidth
- ✓ Scalable, Operando EIS

Learn More ↗

03

Product pages organised around real applications

The Pulse Series page explains the product through the problems it solves: degradation benchmarking, post-field failure analysis, catching stack defects before they ship. Each one links to a detailed Knowledge Centre article, so each product page leads into the detail behind it.

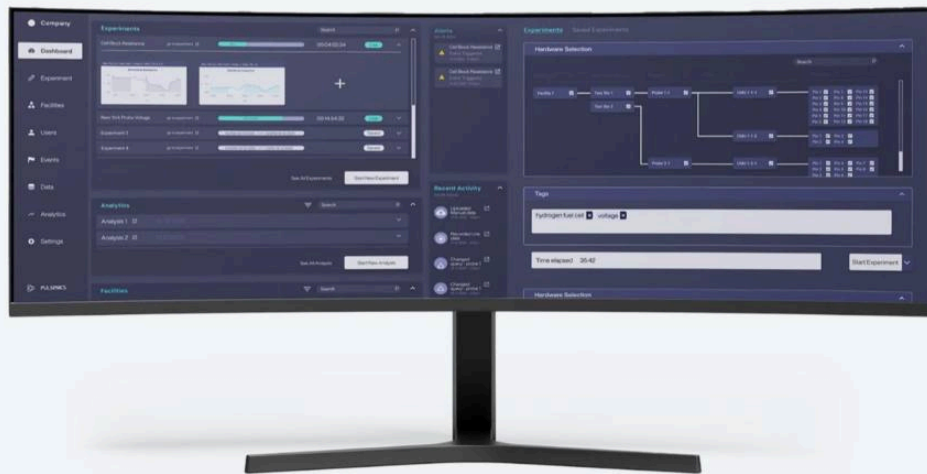


04

Give the software its own section

Every Pulse Series system ships with the Pulse Hub software, so the product page gives it a section of its own, shown on a monitor the way a lab would use it. The interface on that screen is the [System Software I designed with Marc Varone](#).

The Pulse Hub Platform



The Knowledge Centre

I built the Knowledge Centre to improve the way Pulsenics publishes what it knows about EIS. It reads like documentation: how the technology works, then how it's used in battery and hydrogen work. It's all built on the Webflow CMS, so anyone on the team can write and publish articles without touching the layout.

Left

Navigate by topic

Technology basics in order, then applications grouped by industry: battery and hydrogen.

Centre

Read in depth

Long articles with equations, figures and interactive charts inline. The Pulsenics team built the charts; I host them and embed them in the articles.

Right

Jump within a page

An automatic table of contents for each article, plus a link to talk to Pulsenics.

Pulsenics Knowledge Centre

Technology

[Introduction](#)

[1. How does Electrochemical Impedance Spectroscopy work?](#)

[2. Signal Optimization](#)

[3. Interpreting EIS Data](#)

[4. Introduction to Multisine EIS](#)

[References](#)

Applications

Battery

[Grade battery cells from motive and micromobility packs before redeployment](#)

[Grade new incoming cells before integrating into modules or packs](#)

[Grade incoming automotive packs before deployment into stationary storage applications](#)

[Qualify battery modules prior to pack integration](#)

Hydrogen

[Diagnose stack failures and perform post-field forensics](#)

[Detect electrolyser stack defects and prevent quality escapes](#)

Pulsenics Knowledge Centre

Welcome to the Pulsenics Knowledge Centre

This resource serves as a comprehensive guide to understanding all facets of Electrochemical Impedance Spectroscopy (EIS), from the underlying science to real-world applications and measurement best practices.

Whether you're new to impedance analysis or an experienced electrochemist looking to deepen your expertise, this Knowledge Centre is designed to meet you where you are. We've organized content to take you from foundational principles, how and why impedance measurements work, through to advanced topics like equivalent circuit modelling, data interpretation, and troubleshooting common artefacts.

EIS is one of the most powerful and information-rich techniques in electrochemistry, capable of probing processes across many orders of magnitude in frequency. But that power comes with complexity. Our goal here is to make that complexity approachable: clear explanations, practical guidance, and worked examples grounded in the kinds of problems our users actually encounter. Inside, you'll find:

- 1. Science & Theory:** The physics and mathematics of impedance, from basic circuit analogies to the Kramers-Kronig relations
- 2. Instrumentation & Setup:** How to configure your system for reliable, low-noise measurements
- 3. Applications:** How EIS can be used across a range of fields, including batteries, fuel cells, and electrolysers
- 4. Data Analysis:** Fitting strategies, model selection, and common pitfalls
- 5. Best Practices:** Practical tips for getting the most out of every measurement

This is a living resource, updated as the science evolves and as we learn from our community. If you have questions or topics you'd like to see covered, we'd love to hear from you.

[Speak with Pulsenics](#)

Table of Contents

[Welcome to the Pulsenics Knowledge Centre](#)

Knowledge Centre home

Pulsenics Knowledge Centre

Technology

[Introduction](#)

[1. How does Electrochemical Impedance Spectroscopy work?](#)

[2. Signal Optimization](#)

[3. Interpreting EIS Data](#)

[4. Introduction to Multisine EIS](#)

[References](#)

Applications

Battery

[Grade battery cells from motive and micromobility packs before redeployment](#)

[Grade new incoming cells before integrating into modules or packs](#)

[Grade incoming automotive packs before deployment into stationary storage applications](#)

[Qualify battery modules prior to pack integration](#)

Hydrogen

[Diagnose stack failures and perform post-field forensics](#)

[Detect electrolyser stack defects and prevent quality escapes](#)

3. Interpreting EIS Data

3.0.1 The Nyquist Plot

As previously mentioned in Module 1, the Nyquist plot is one of the most common ways to quickly visualize and interpret raw EIS data. Based on the type of electrochemical system and its state, the shape of the Nyquist plot may differ. As temperature, state of charge, and health change, we can observe the differences in the Nyquist data. The interactive figure below shows the effect of state of charge on the Nyquist plot's signature.

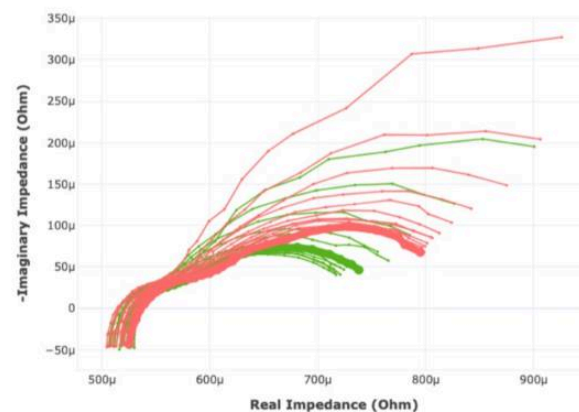


Table of Contents

[3.0.1 The Nyquist Plot](#)

[3.0.2 Distribution of Relaxation Times](#)

[3.0.3 Example Equivalent Circuit Model Process](#)

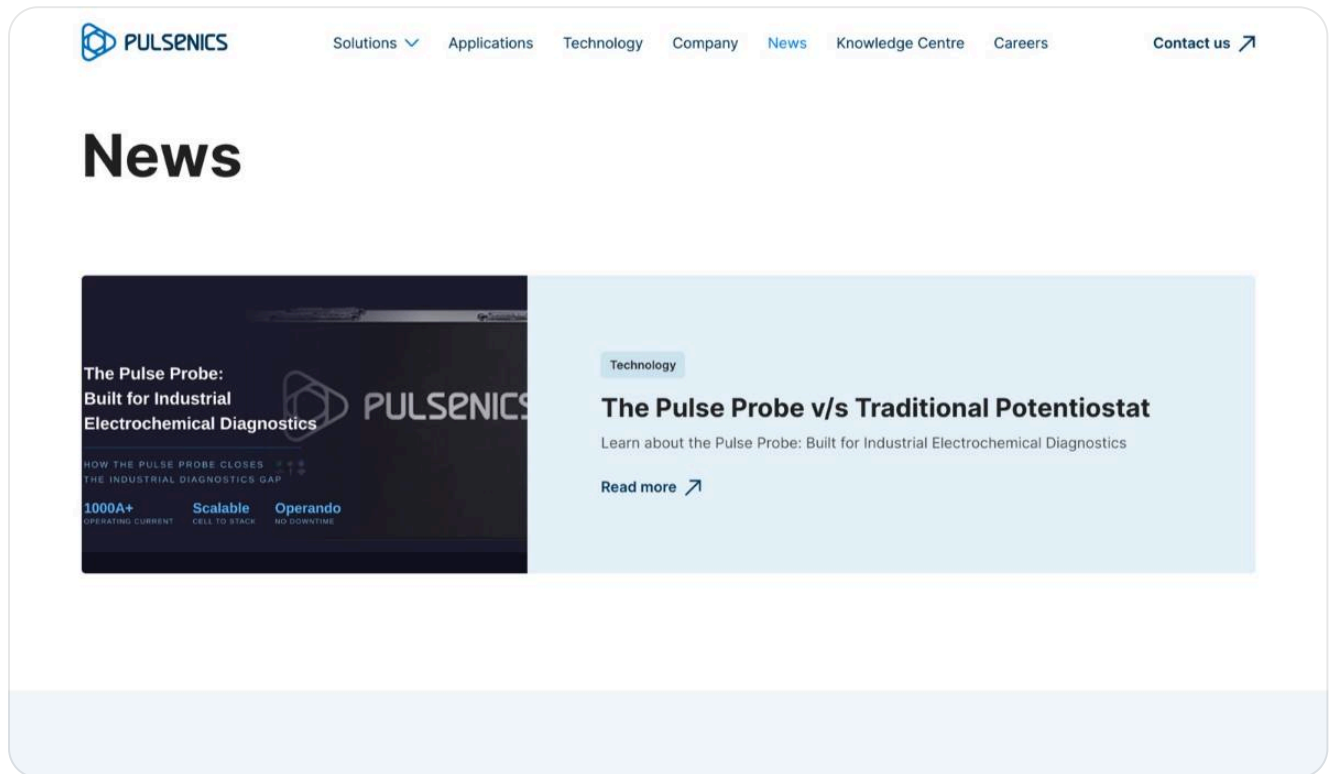
[Want to learn more? Speak with us today!](#)

An article with an interactive chart

News

Pulsenics publishes partnerships, conference talks, product launches and studies almost every month. The News page has a featured story on top and a filterable grid underneath: Announcement, Conference, Technology, Company, Collaboration, and Studies & Papers.

It runs on the CMS like the Knowledge Centre. Someone on the team adds a story, and it shows up on the News page and in the latest news banner at the top of the homepage. Nobody needs me to keep the site current.



Featured story

Latest

Filter by:

[Announcement](#)

[Conference](#)

[Technology](#)

[Company](#)

[Collaboration](#)

[Studies & Papers](#)

[Clear filters](#)



Conference

Pulsenics shares our learnings from Energy Storage Canada 2026

Incoming cell quality control remains critical for this fast-growing industry.

[Read more](#) ↗



Technology

The Pulse Probe v/s Traditional Potentiostat

Learn about the Pulse Probe: Built for Industrial Electrochemical Diagnostics

[Read more](#) ↗



Company

Pulsenics exhibits at Canada Investment Summit 2026

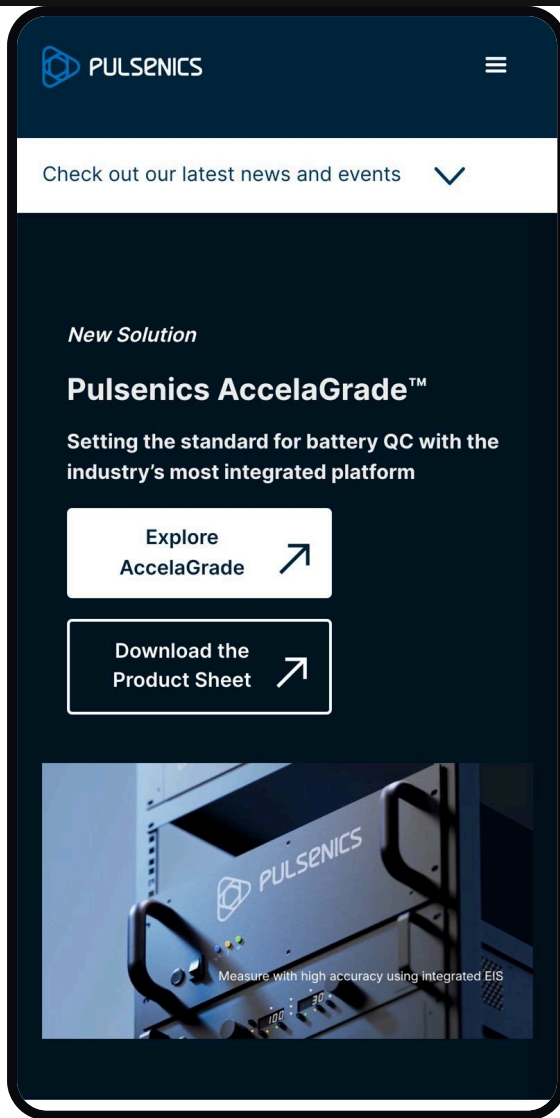
High-profile gathering hosted by Prime Minister Mark Carney and Ontario Premier Doug Ford brought global leaders to the Royal Ontario Museum.

[Read more](#) ↗

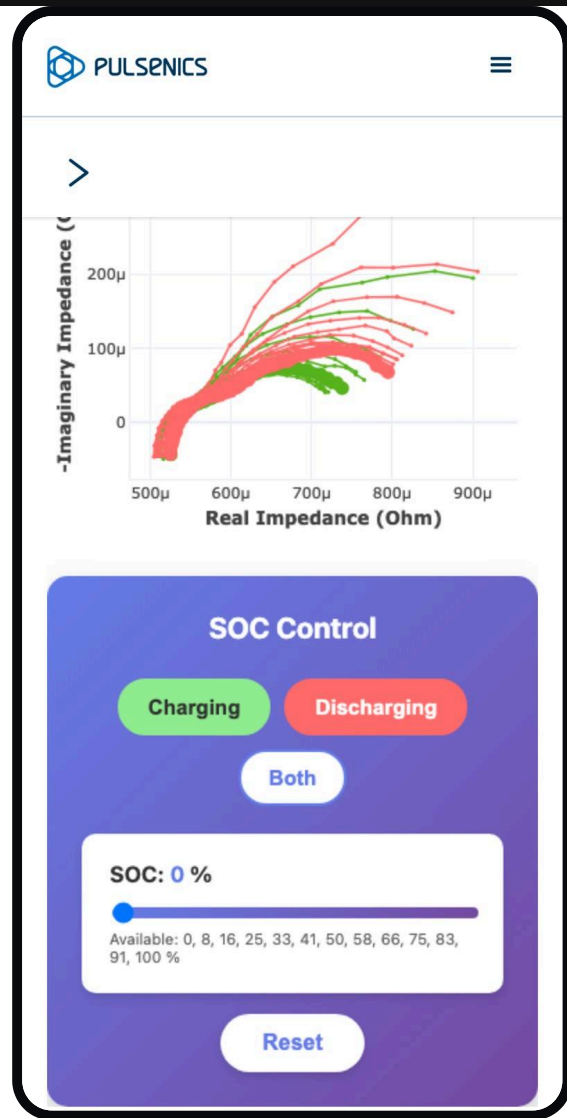
Category filters and the article grid

On a phone

Every page is responsive, including the articles and the charts inside them.



Homepage. Actions stack full width.



Articles. Embedded charts still work on a phone.

See it live

Live, and updated all the time

The team keeps it updated with news and Knowledge Centre content, and I keep building on it as Pulsenics launches products and publishes research.